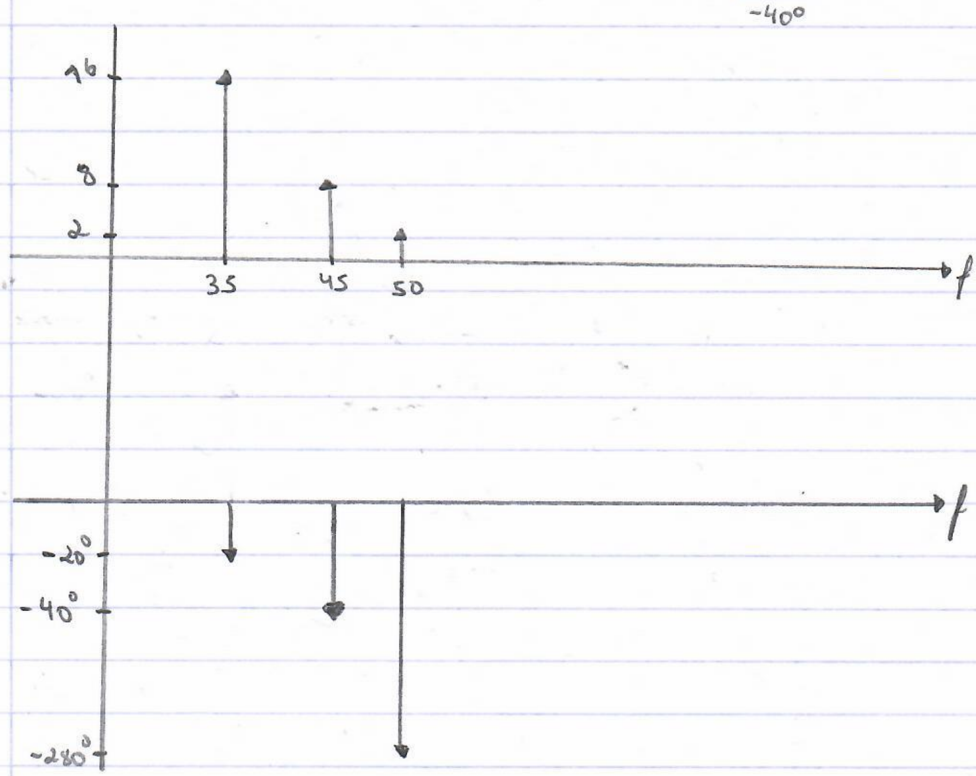
TPC

1. $v(t) = 16 . \cos(70\pi t - 20^\circ) - 8 . \mathrm{sen}(90\pi t + 50^\circ) - 2 . \mathrm{sen}(100\pi t - 10^\circ)$

$\Leftrightarrow v(t) = 16 . \cos(2\pi 35t - 20^\circ) + 8\cos(2\pi 45t + \underbrace{50^\circ - 90^\circ}_{-40^\circ}) + 2\cos(2\pi 50t \underbrace{- 10^\circ - 90^\circ - 180^\circ}_{-280^\circ})$



$$v(t) = \frac{16}{2}\left(e^{-j20} \times e^{j2\pi 35t} + e^{j20} \times e^{j2\pi 35t}\right) + \frac{8}{2}\left(e^{-j40} \times e^{j2\pi 45t} + e^{j40} \times e^{-j2\pi 45t}\right) +$$

$$\frac{2}{2}\left(e^{-j280} \times e^{j2\pi 50t} + e^{j280} \times e^{-j2\pi 50t}\right)$$